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ExpenseVoyage

ExpenseVoyage

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CERTIFICATE

OF APPRECIATION

This certificate is proudly presented to :

Falak, Mahnoor, Nadia, Aakash

Have successfully Designed and Developed

ExpenseVoyage

August 14,2025

Internship Supervisor

Problem Definition

Traveling, whether for personal leisure or business purposes, often involves managing multiple expenses across different categories such as transportation, accommodation, food, and activities. In most cases, travelers face challenges in tracking and organizing these expenses efficiently. Traditional methods, such as manually entering data into spreadsheets or storing paper receipts, are time-consuming, error-prone, and lack real-time accessibility.

Additionally, when traveling in groups, managing shared expenses becomes even more complicated. There is no easy way to split costs, keep all members updated, or ensure transparency in spending. Furthermore, many travelers struggle with staying within budget due to the absence of timely alerts and organized tracking tools.

The lack of a centralized, user-friendly, and accessible platform for **real-time expense tracking, trip planning, and budget management** results in overspending, financial mismanagement, loss of important data, and difficulty in reimbursement claims.

With the increasing use of smartphones and internet accessibility, there is a strong need for a **web-based application** that can serve as a single platform for trip planning, itinerary management, expense logging, currency conversion, and budget monitoring. Such a system should be secure, scalable, easy to use, and provide real-time updates and notifications to assist travelers in managing their journeys more efficiently.

ExpenseVoyage aims to solve these problems by providing a **comprehensive, collaborative, and user-friendly travel expense management system** for both individual and group travelers.

Customer Requirement Specification

The customers (users) of the Expense Management Application expect a comprehensive and reliable platform to assist in managing their financial transactions and budgets.

Functional Requirements:

User Features

1. **Home Page & Dashboard**

- Personalized welcome message upon login.
- Overview of current/upcoming trips, destinations, and remaining budgets.
- List of recent activities (expenses logged, itinerary updates).

2. **User Management**

- Account registration and login.
- Password recovery and reset.
- Profile management with preferred currency and travel preferences.
- Profile picture upload.

3. **Trip Planning**

- Create, edit, delete trips with destination, start, and end dates.
- Add itinerary items (flights, hotels, activities).
- Categorize itinerary items (e.g., transport, lodging, activities).

4. **Expense Tracking**

- Manual logging of expenses (amount, category, date, notes).
- Edit/delete expenses.
- Set overall and category-specific budgets.
- Real-time budget tracking with alerts.

5. **Currency Conversion**

- Support for at least USD, EUR, GBP, and local currency.
- Convert all expenses into preferred currency with current exchange rates.

6. **Reports**

- Generate expense reports by trip and category.

7. **Notifications & Reminders**

- Daily reminders to log expenses.
- Alerts for upcoming itinerary items.
- Budget threshold notifications.

Admin Features

- Admin login.
- Manage destinations, trips, currencies, and exchange rates.

Common Features (User & Admin)

- Search/filter trips or destinations.
- Sort results by specific criteria.
- “About Us” section with service details.
- Image gallery of destinations.
- “Contact Us” form with name, email, contact number, and message fields.
- GPS functionality where applicable.

Non-Functional Requirements:

- Ease of Use: Clear navigation, intuitive menus.
- Performance: Fast loading times and smooth navigation.
- Accessibility: Legible fonts, color contrast, and mobile responsiveness.
- Security: Secure authentication, encrypted data storage.
- Availability: 24/7 uptime with minimal downtime.
- Compatibility: Support for modern browsers and devices.

- Scalability: Handle growing user base and feature expansion.

Hardware

- A minimum computer system that will help you access all the tools in the courses is a Pentium 166 or better
- 128 Megabytes of RAM or better
- Windows 2000 Server (or higher if possible)

Software

- Use software as per your requirement
- Windows OS /MongoDB/Express/React/Node.js/Notepad

Packages

This packages are used in express project

"bcrypt": "^6.0.0",

"body-parser": "^2.2.0",

"cors": "^2.8.5",

"express": "^5.1.0",

"mongoose": "^8.17.1",

"multer": "^2.0.2",

"nodemon": "^3.1.10"

This packages are used in React project

"@fortawesome/fontawesome-free": "^7.0.0",

"@popperjs/core": "^2.11.8",

"axios": "^1.11.0",

"bootstrap": "^5.3.7",

"ionicons": "^8.0.13",

"jquery": "^3.7.1",

"jspdf": "^3.0.1",

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"jspdf-autotable": "^5.0.2",
"leaflet": "^1.9.4",
"leaflet-defaulticon-compatibility": "^0.1.2",
"react": "^19.1.1",
"react-dom": "^19.1.1",
"react-icons": "^5.5.0",
"react-router-dom": "^7.8.0",
"react-toastify": "^11.0.5",
"recharts": "^3.1.2",
"sass": "^1.90.0",
"simple-line-icons": "^2.5.5",
"slick-carousel": "^1.8.1"
```

Scope Of Work

The **ExpenseVoyage** Web Application aims to provide travelers—both individuals and groups—with a comprehensive, intuitive, and collaborative platform for managing trip planning and expense tracking. The system will streamline the process of organizing itineraries, logging expenses, managing budgets, and generating reports, all within a secure and responsive web interface.

The scope of this project includes:

User Management

- User registration, login, and profile management with profile picture upload.
- Password recovery and reset functionality.

Trip Planning

- Create, view, edit, and delete trips with destination, dates, and budget.
- Itinerary management including categorization (transport, lodging, activities, etc.).

Expense Tracking

- Manual logging of expenses with amount, category, date, and notes.
- Budget management with alerts for overspending.
- Multi-currency support with real-time conversion.

Reporting

- Detailed expense reports categorized by trip and expense type.

Notifications & Reminders

- Alerts for budget limits and reminders for itinerary items or expense logging.

Administrative Functions

- Admin management of destinations, trips, currencies, and exchange rates.

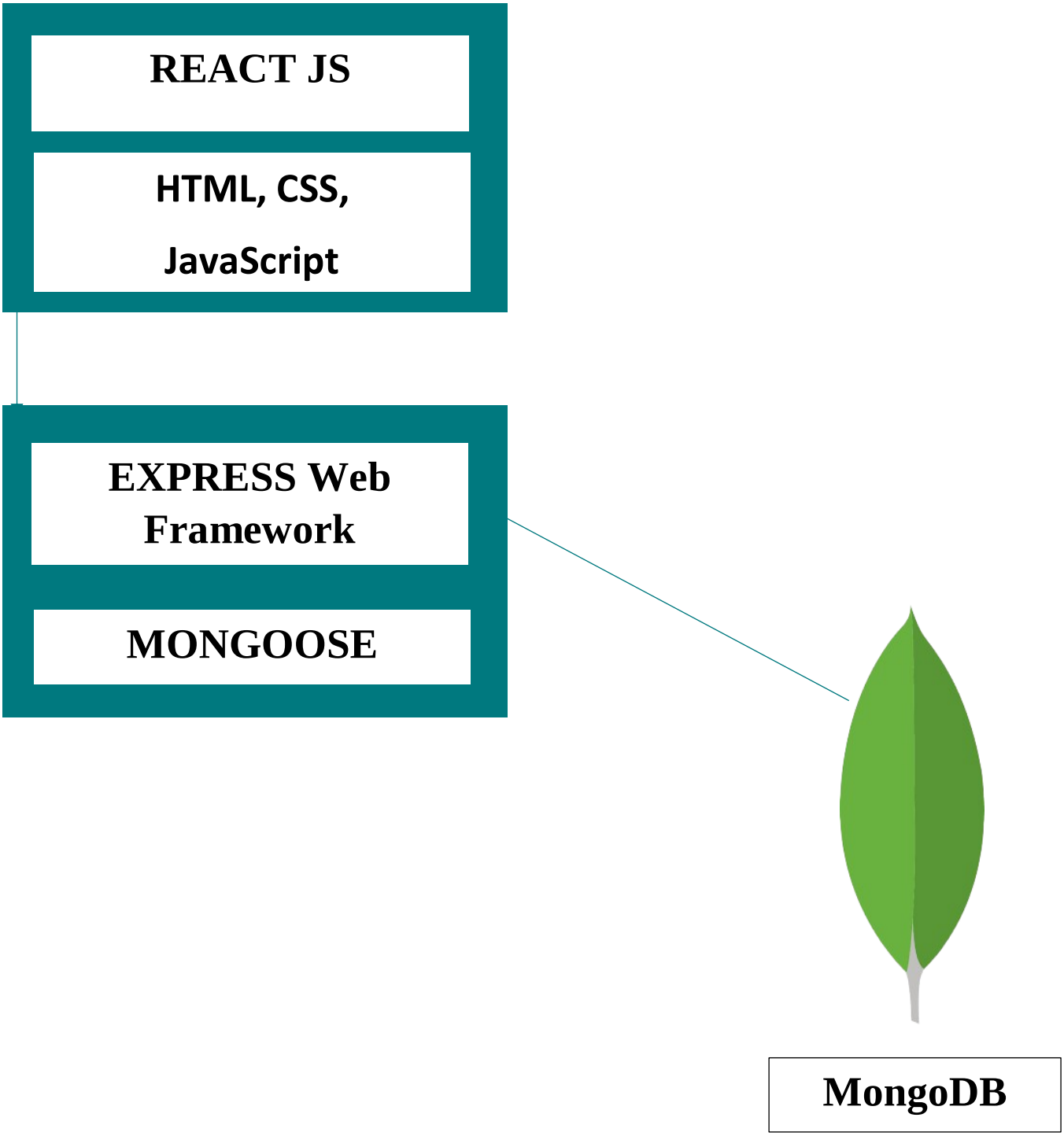
General Features

- Search, filter, and sort functionalities for trips, expenses, and itineraries.
- About Us, Image Gallery, and Contact Us pages.
- GPS integration for location-based features.

Non-Functional Goals

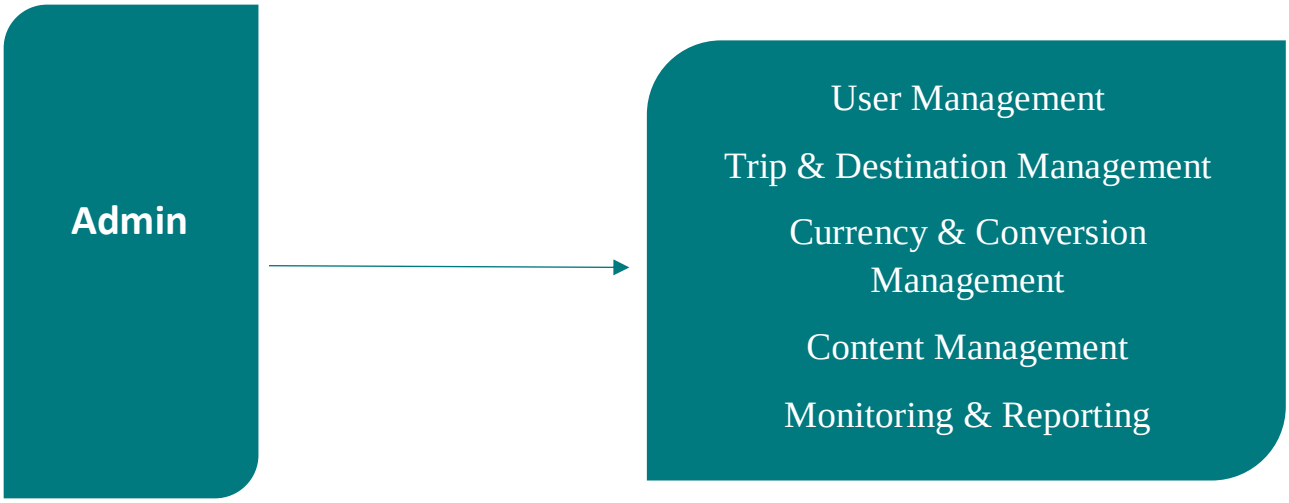
- Cross-browser compatibility and responsive design for various devices.
- Secure authentication and data handling.
- High performance with minimal load times.
- Scalability to handle increased data and user load.

Architecture And Design Of The Project

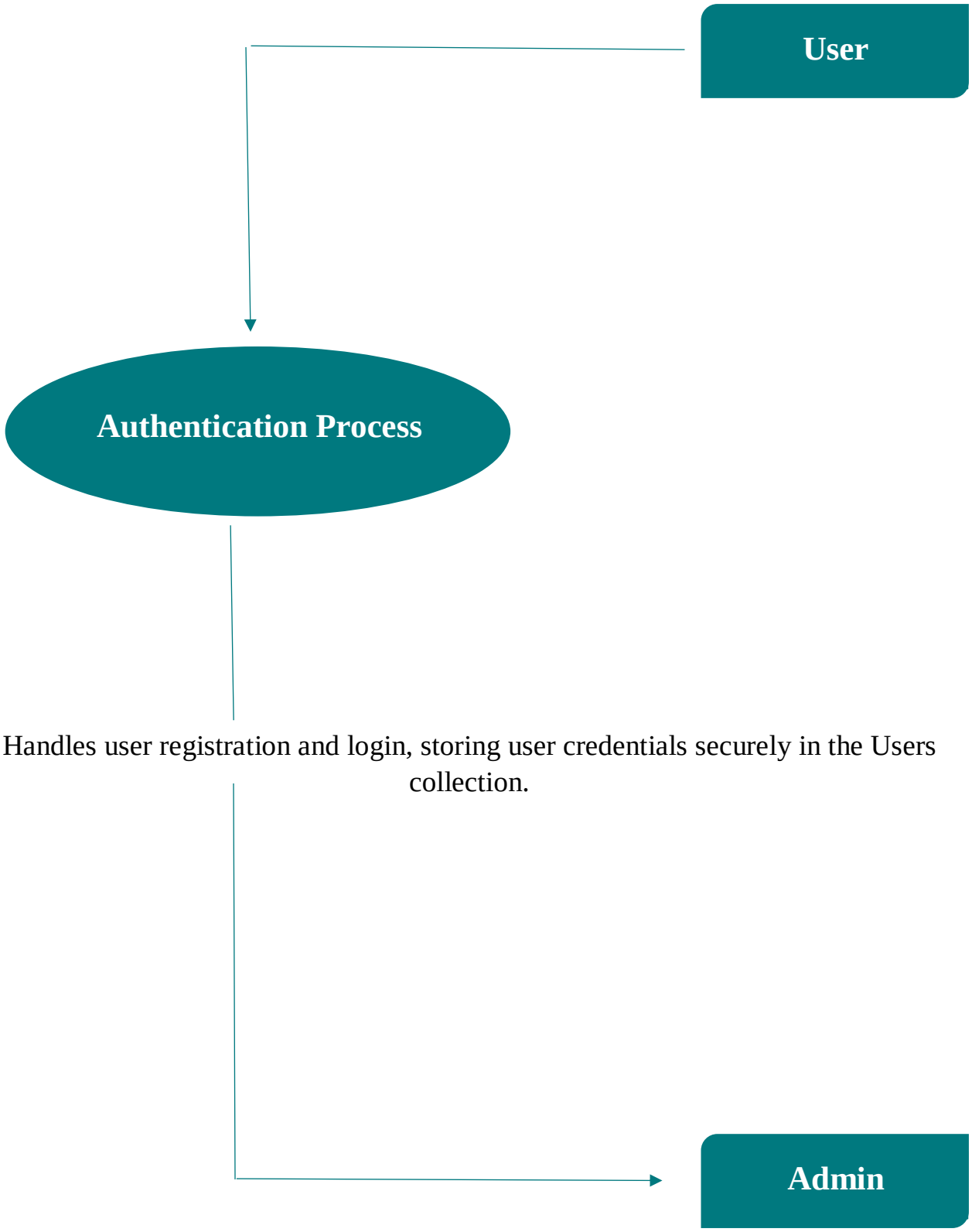


Application Architecture

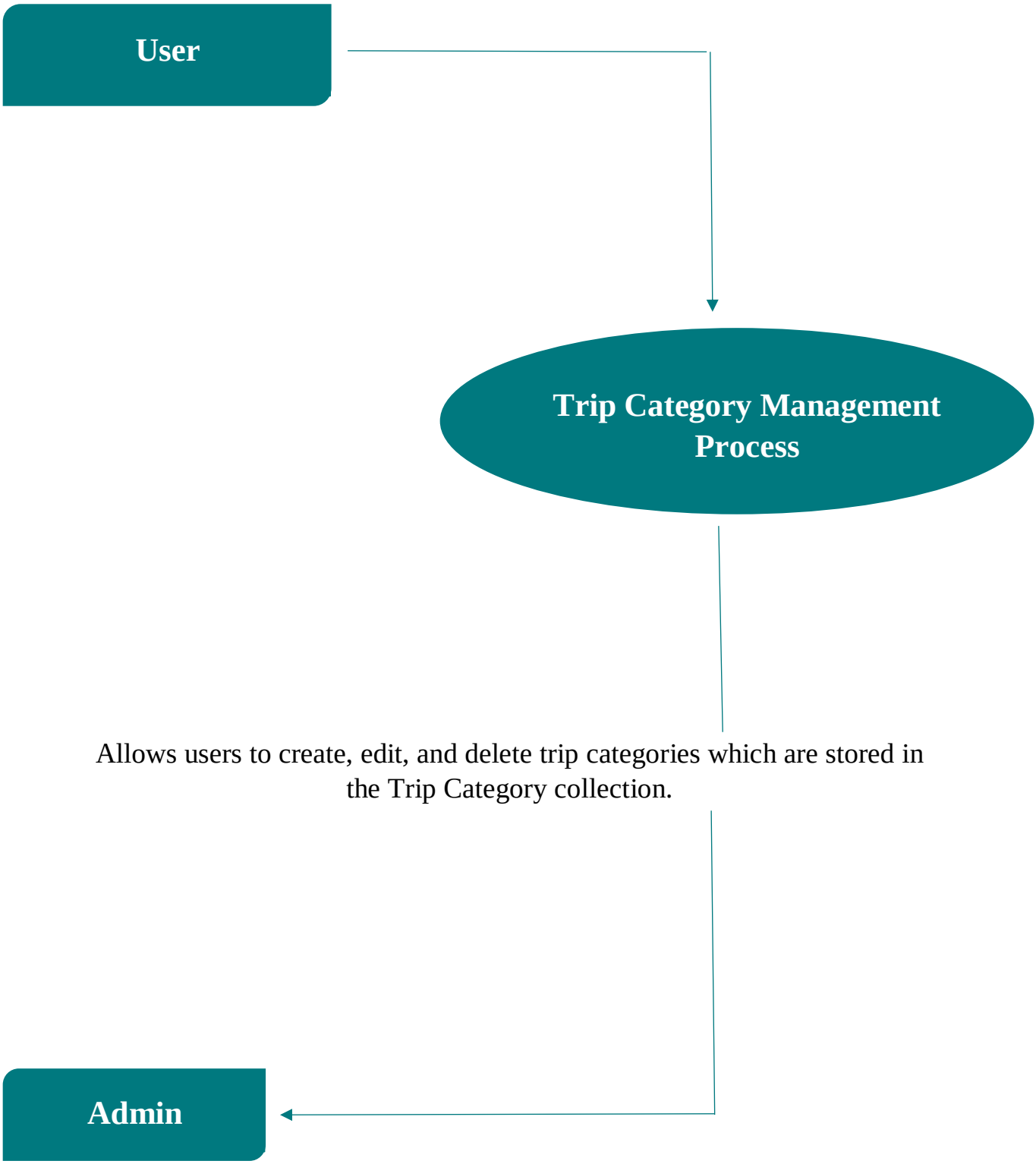
Data Flow Diagram



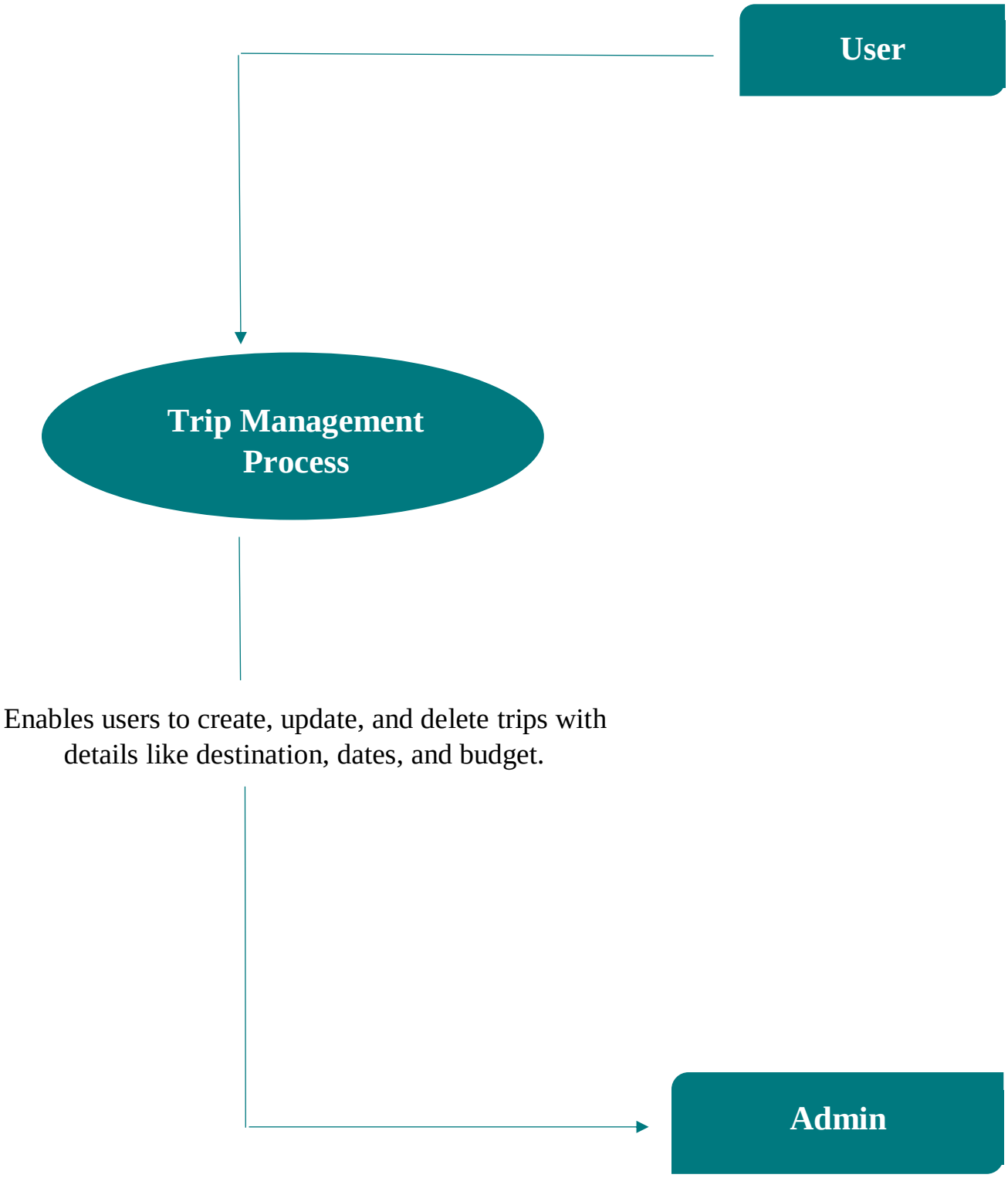
Dfd For User Authentication



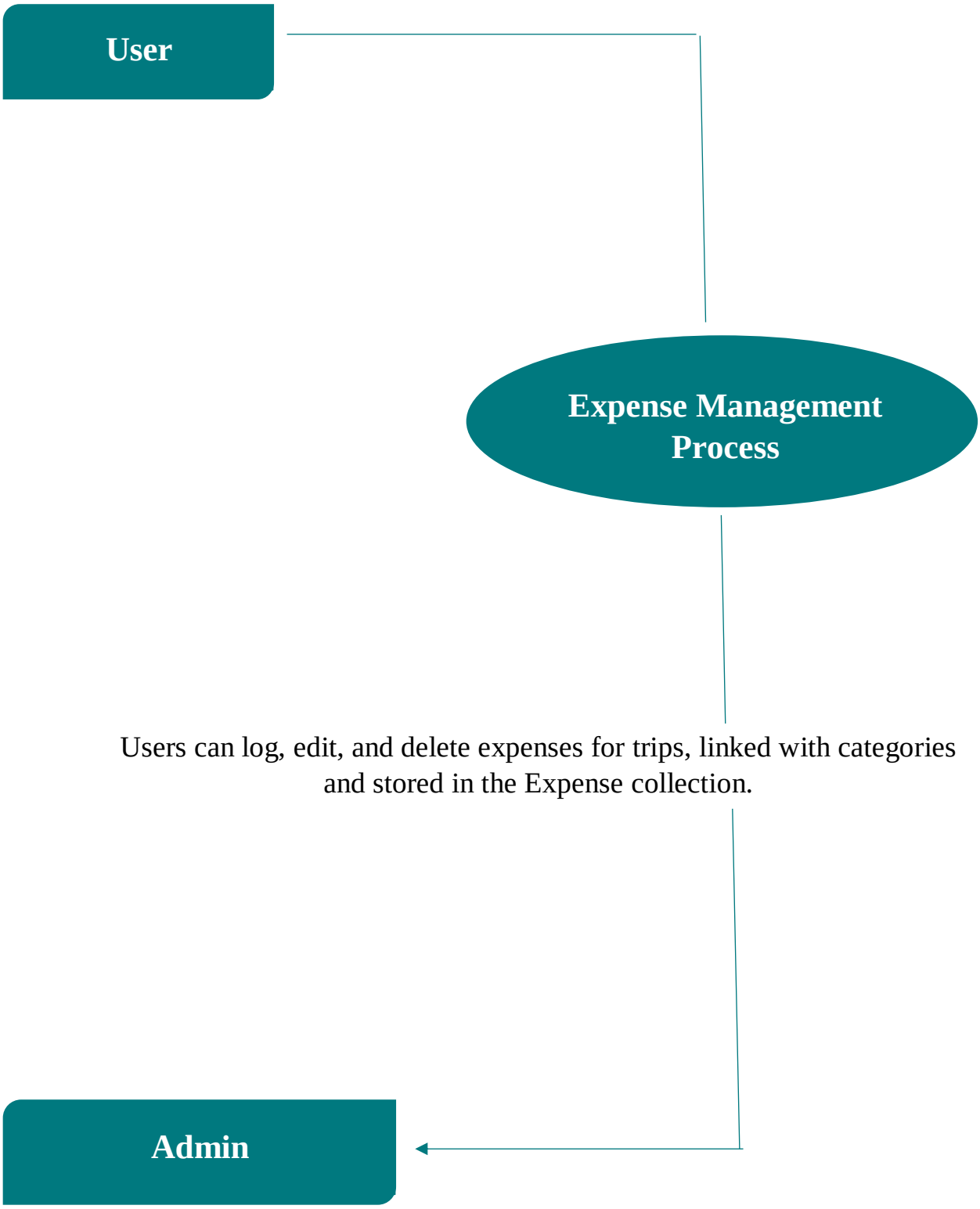
DFD for Trip Category Management



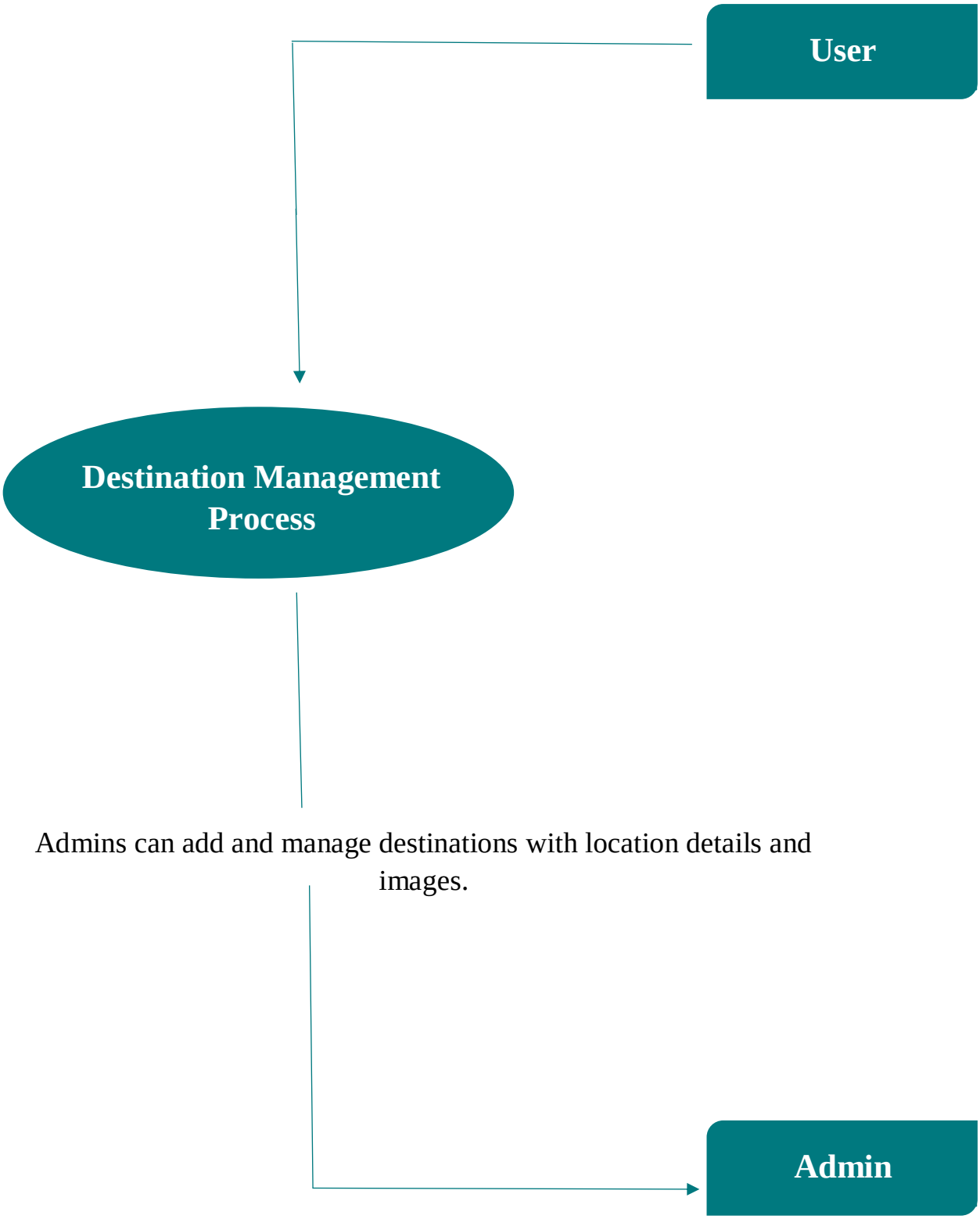
DFD for Trip Management



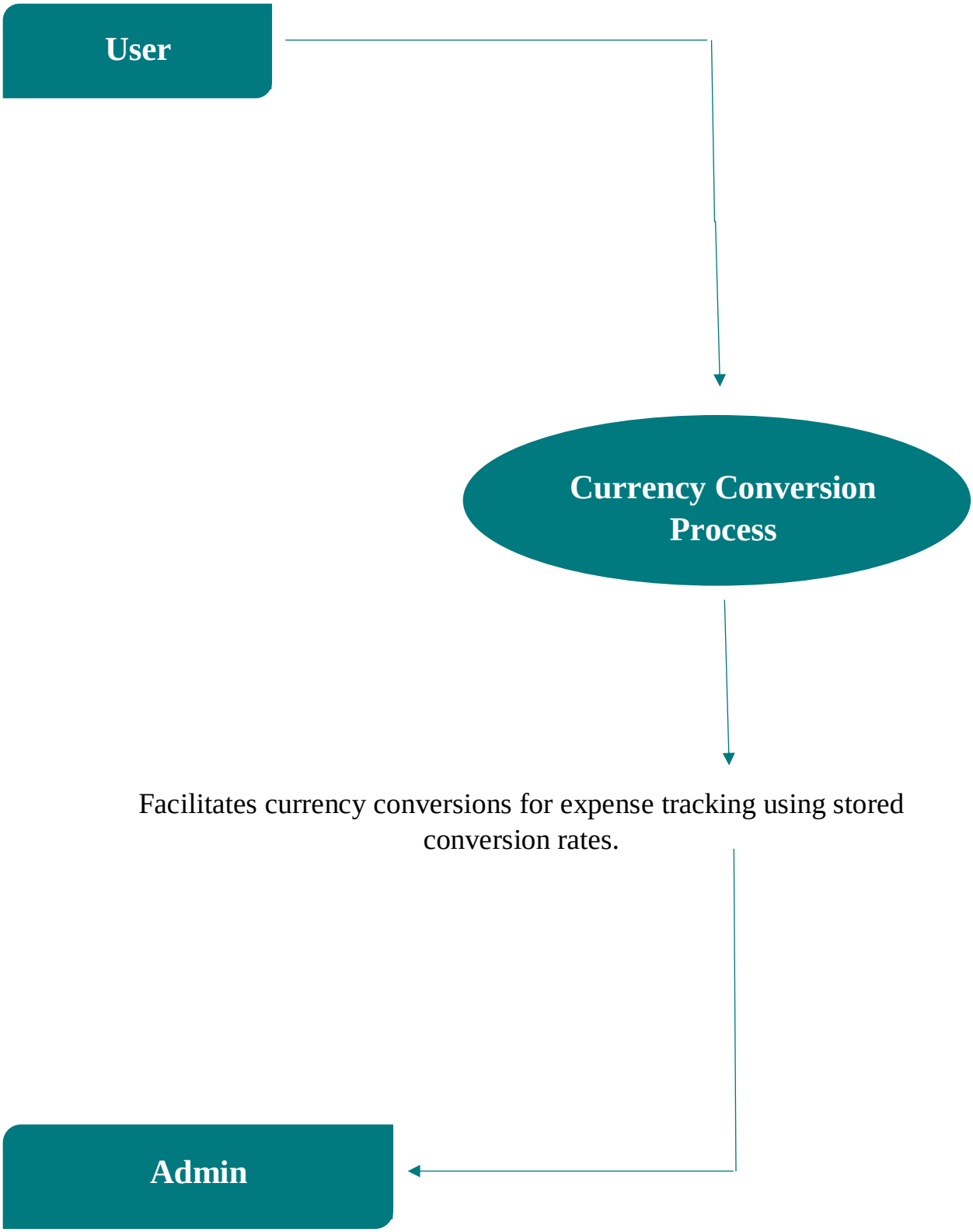
DFD for Expense Management



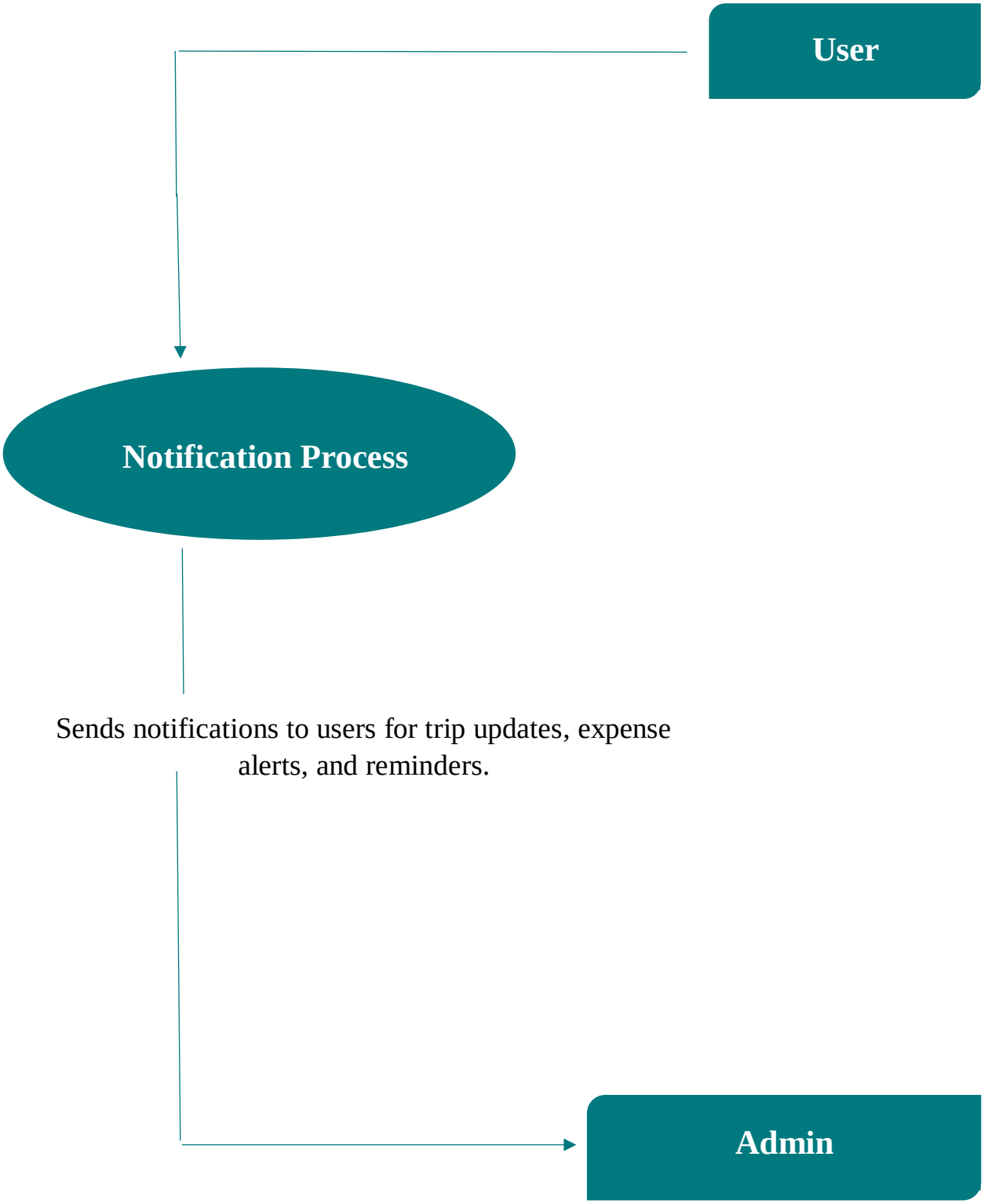
DFD for Destination Management



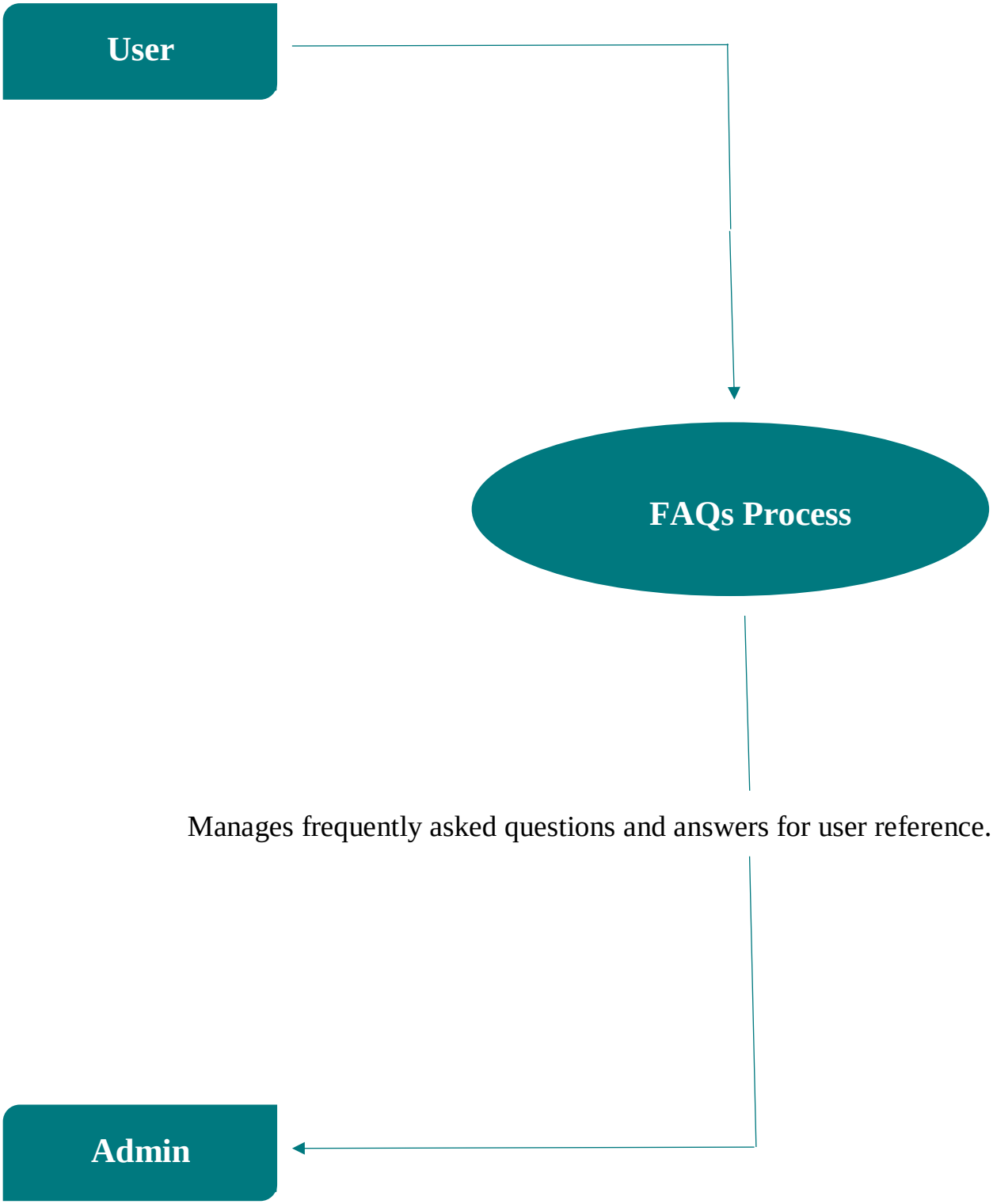
DFD for Currency Conversion



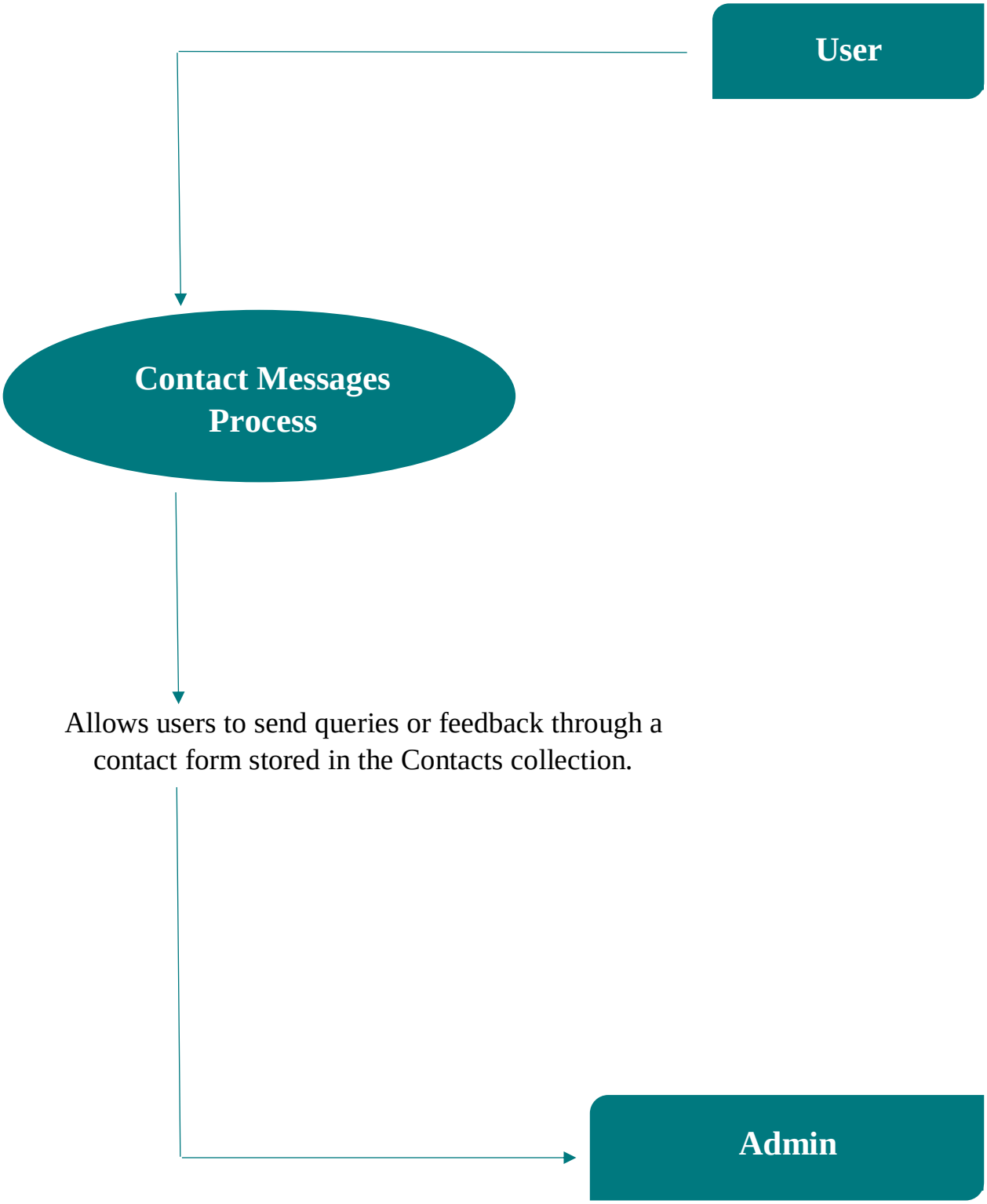
DFD for Notifications



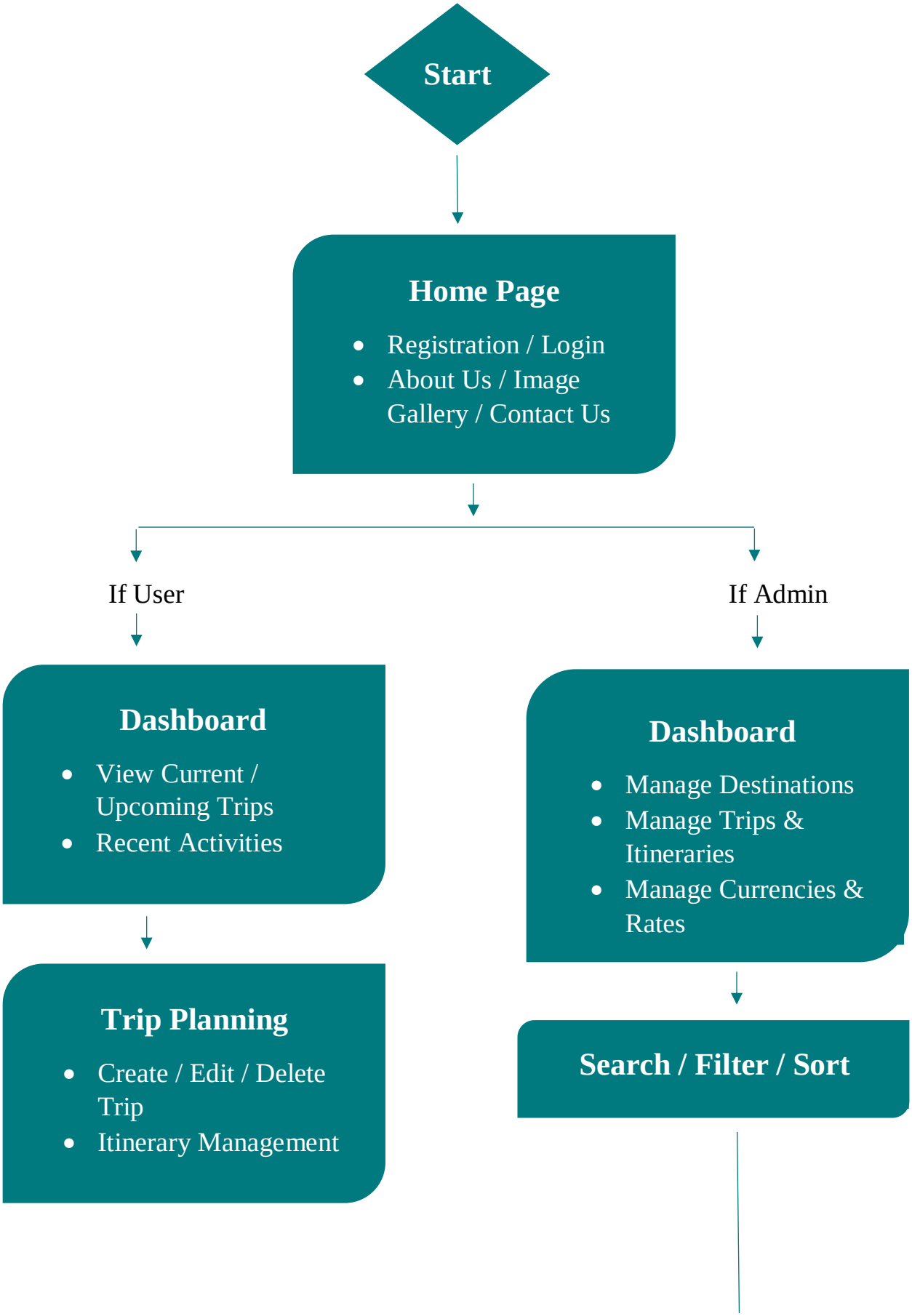
DFD for FAQs

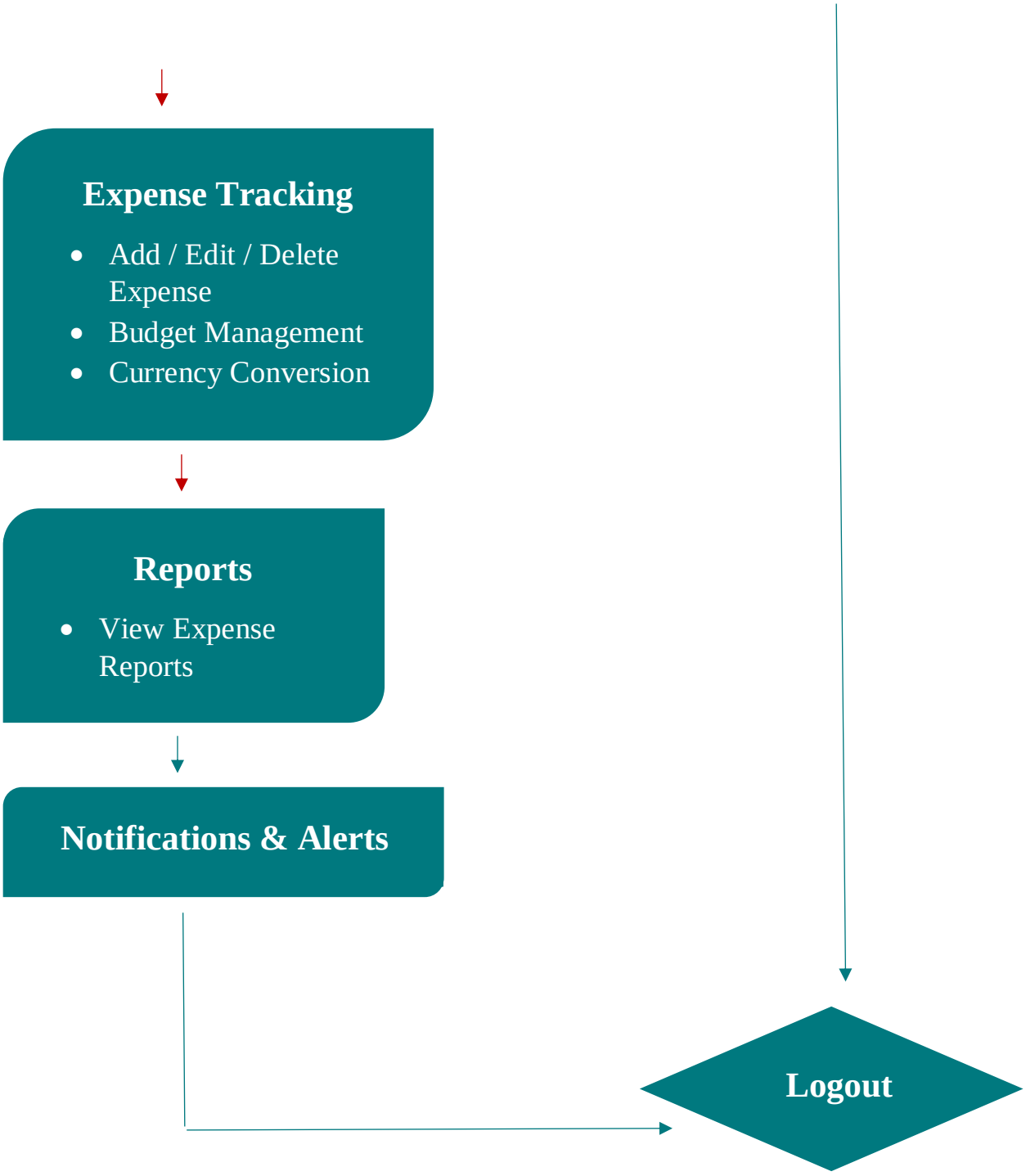

User
FAQs Process
Manages frequently asked questions and answers for user reference.
Admin

DFD for Contact Messages



Flowchart


Start
Home Page
Registration / Login
About Us / Image Gallery / Contact Us
If User
If Admin
Dashboard
View Current / Upcoming Trips
Recent Activities
Trip Planning
Create / Edit / Delete Trip
Itinerary Management
Dashboard
Manage Destinations
Manage Trips & Itineraries
Manage Currencies & Rates
Search / Filter / Sort


Expense Tracking
Add / Edit / Delete Expense
Budget Management
Currency Conversion
Reports
View Expense Reports
Notifications & Alerts
Logout

Database Design And Structure

User

```
{
  _id: ObjectId,
  first_name: String,
  last_name: String,
  email: String,
  password: String,
  profile_picture: String,
  status: Number,
  is_admin: Number (0 or 1)
}
```

Trip Category

```
{
  _id: ObjectId,
  user_id: String,
  category_name: String,
  status: Number,
}
```

Trip

```
{
  _id: ObjectId,
  user_id: String,  // Should be ObjectId, ref: User
  trip_name: String,
  start_date: Date,
  end_date: Date,
  destination: String,
  budget: Number,
  accomodation: Number,
  food: Number,
  transport: Number,
  preferred_currency: String,
}
```

Expense

```
{
  _id: ObjectId,
  user_id: String,  // Should be ObjectId, ref: User
  trip_name: String,
  category_name: String,
```

```
  amount: Number,
  expense_date: Date,
  notes: String,
}
```

Destination

```
{
  _id: ObjectId,
  User_id: String,
  name: String,
  description: String,
  image_url: String,
  status: Number,
  longitude: Number,
  latitude: Number,
}
```

Conversions

```
{
_id: ObjectId,
userId: String
fromCurrency: String
toCurrency: String
```

amount: Number

convertedAmount: Number

rate: Number

date: Date

}

Notifications

{

_id: ObjectId,

user_id: ObjectId (ref: User),

message: String,

status: String,

}

FAQ’s

{

_id: ObjectId,

subject: String,

Faq_answer: String,

Faq_question: String,

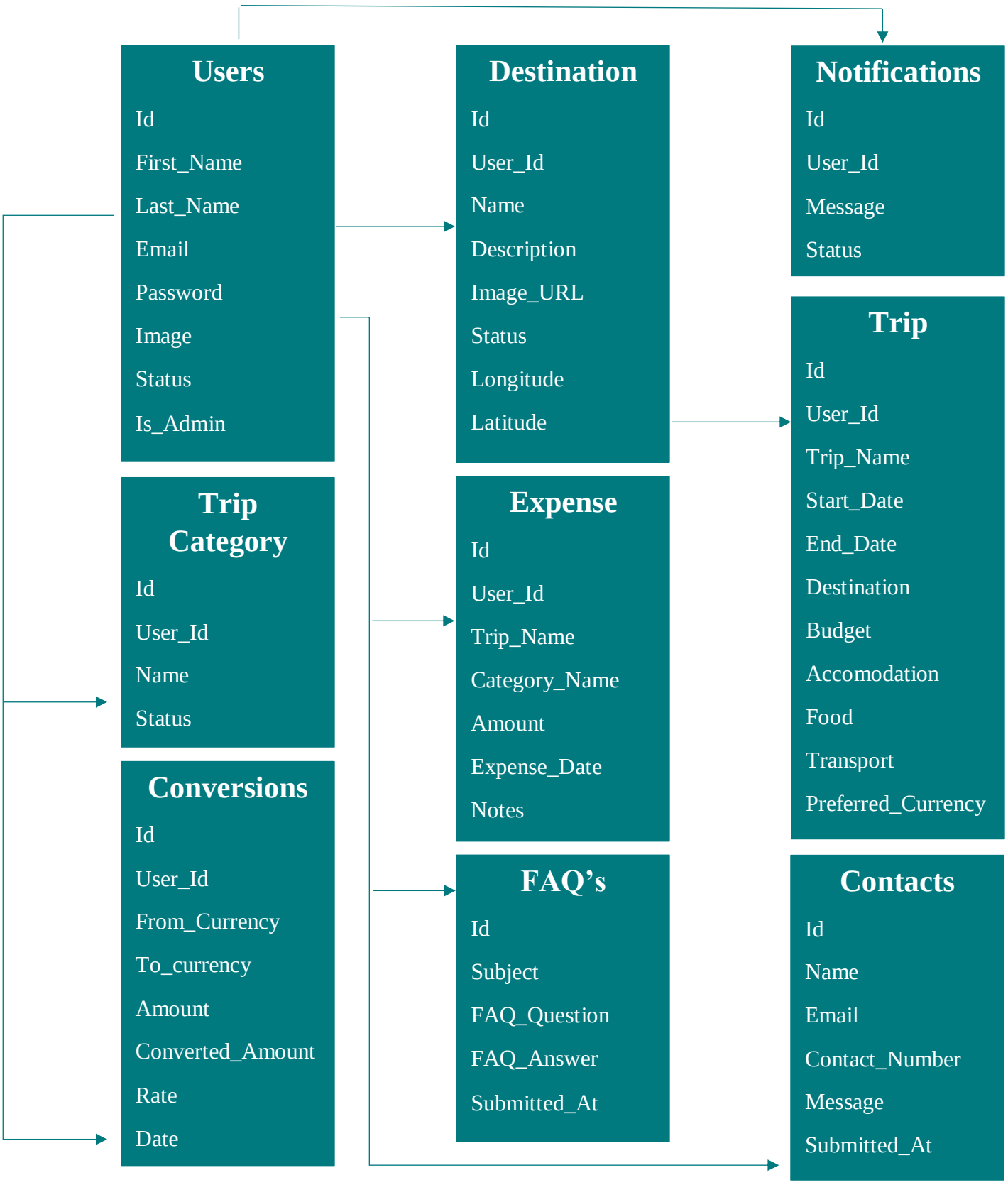
submitted_at:Date,

}

Contacts

```
{
_id: ObjectId,
name: String,
email: String,
contact_number: Number,
message: String,
 submitted_at:Date,
}
```

Entity Relationship (ER) Diagrams



Task Sheet

1. Research & Planning

Task ID	Task Description	Responsible	Status
01	Study SRS document and finalize feature list	Mahnoor, Falak, Nadia, Aakash	Completed
02	Define tech stack (Frontend, Backend, Database)	Mahnoor, Falak, Nadia, Aakash	Completed
03	Identify constraints and licensing requirements for images/videos	Mahnoor, Falak, Nadia, Aakash	Completed

2. UI/UX Design

Task ID	Task Description	Responsible	Status
01	Create wireframes for all pages (Home, Dashboard,	Mahnoor, Falak, Nadia, Aakash	Completed

	Trip, Expense, Reports, etc.)		
02	Design responsive layouts for cross-device compatibility	Mahnoor, Falak, Nadia, Aakash	Completed
03	Develop sitemap and navigation flow	Mahnoor, Falak, Nadia, Aakash	Completed
04	Prepare "About Us", "Contact Us", and Image Gallery mockups	Mahnoor, Falak, Nadia, Aakash	Completed

3. Frontend Development

Task ID	Task Description	Responsible	Status
01	Implement Home Page with personalized welcome message	Mahnoor, Falak, Nadia, Aakash	Completed
02	Dashboard showing current/upcoming trips and recent activities	Mahnoor, Falak, Nadia, Aakash	Completed
03	Trip Planning pages (Create/Edit/Delete/View trips)	Mahnoor, Falak, Nadia, Aakash	Completed

04	Itinerary Management module (categorization, CRUD operations)	Mahnoor, Falak, Nadia, Aakash	Completed
05	Expense Logging pages (manual entry, category, date, notes)	Mahnoor, Falak, Nadia, Aakash	Completed
06	Budget tracking with alerts and category-specific budgets	Mahnoor, Falak, Nadia, Aakash	Completed
07	Multi-currency support with conversion UI	Mahnoor, Falak, Nadia, Aakash	Completed
08	Reports page for expense summaries	Mahnoor, Falak, Nadia, Aakash	Completed
09	Notifications & reminders UI integration	Mahnoor, Falak, Nadia, Aakash	Completed
10	Search, filter, and sort features for trips & expenses	Mahnoor, Falak, Nadia, Aakash	Completed

4. Backend Development

Task ID	Task Description	Responsible	Status
01	User authentication (registration, login,	Mahnoor, Falak, Nadia, Aakash	Completed

	password recovery)		
02	User profile management with preferred currency & image upload	Mahnoor, Falak, Nadia, Aakash	Completed
03	Trip management API endpoints	Mahnoor, Falak, Nadia, Aakash	Completed
04	Itinerary CRUD API endpoints	Mahnoor, Falak, Nadia, Aakash	Completed
05	Expense logging & budget management API	Mahnoor, Falak, Nadia, Aakash	Completed
06	Currency conversion service integration	Mahnoor, Falak, Nadia, Aakash	Completed
07	Report generation API	Mahnoor, Falak, Nadia, Aakash	Completed
08	Notification & reminder backend logic	Mahnoor, Falak, Nadia, Aakash	Completed
09	Admin panel for destinations,	Mahnoor, Falak, Nadia, Aakash	Completed

	trips, and currency rates		
10	Search, filter, sort API implementation	Mahnoor, Falak, Nadia, Aakash	Completed

5. Database Development

Task ID	Task Description	Responsible	Status
01	Design database schema (Users, Trips, Expenses, Categories, etc.)	Mahnoor, Falak, Nadia, Aakash	Completed
02	Create and optimize MongoDB Collections	Mahnoor, Falak, Nadia, Aakash	Completed
03	Seed test data for development	Mahnoor, Falak, Nadia, Aakash	Completed

6. Testing

Task ID	Task Description	Responsible	Status
01	Unit testing for backend APIs	Mahnoor, Falak, Nadia, Aakash	Completed

02	Frontend functionality testing (UI/UX flow)	Mahnoor, Falak, Nadia, Aakash	Completed
03	Cross-browser compatibility testing	Mahnoor, Falak, Nadia, Aakash	Completed
04	Responsive design testing on multiple devices	Mahnoor, Falak, Nadia, Aakash	Completed
05	Performance & load testing	Mahnoor, Falak, Nadia, Aakash	Completed
06	Security testing (authentication, access control)	Mahnoor, Falak, Nadia, Aakash	Completed

7. Documentation

Task ID	**Task Description**	**Responsible**	**Status**
01	Write complete project report as per SRS requirements	Falak	Completed
02	Prepare user manual for Admin & Users	Falak	Completed
03	Create demo video	Mahnoor	Completed

	showcasing all functionalities		

Acknowledgement

We would like to acknowledge all those who have given moral support and helped us make the project a success. We wish to express our gratitude to ***Miss Laraib Qadeer***, who guided and helped us. We would also like to express our gratitude to all the staff members of our center for not only providing us with the opportunity to work with them on this project, but also for their support and encouragement throughout the process. We also express our sincere gratitude to our project guide at the organization, for her valuable guidance and support for the completion of this project. And finally, We would like to offer many thanks to all our colleagues for their valuable suggestions and constructive feedback.